

7.6.8. Manage Privilege

An element with the Manage privilege defined SHALL support Read (if readable), Write (if writable), and Invoke (if invocable) access if the source of the request is granted the Manage privilege.

7.6.9. Administer Privilege

An element with the Administer privilege defined SHALL support Read (if readable), Write (if writable), and Invoke (if invocable) access if the source of the request is granted the Administer privilege.

7.6.10. Timed Interaction

This access modifier requires the use of a timed interaction.

Timed interactions are used to limit the amount of time an action message is valid and can interact with a node. They are used to prevent a timing attack on the system. For example, a malicious attacker could perform an "intercept, interfere, and replay" procedure whereby a legitimate message is intercepted, receipt by the intended destination is jammed, and the attacker sends the message at a later time to cause a malicious action such as unlocking a door at an unintended time. While the practical difficulties of such an attack are high, and the malicious eavesdropper cannot decrypt the action message, the timed interaction provides further mitigation of risk for critical actions.

The timed interaction can be thought of as a 2-phase commit. A precursor action ([Timed Request Action](#)) is sent to indicate the valid time window for arrival of some subsequent, primary action. Since the timed request requires a response, an attacker cannot do the store-and-forward timing attack anymore. The lack of an authenticated response from the intended destination will prevent the subsequent primary action from being sent.

A command with this modifier SHALL require a timed invoke interaction. A writable attribute with this modifier SHALL require a timed write interaction.

An attempted untimed write interaction to a writable attribute with this modifier SHALL generate an error response.

An untimed invoke interaction for a request command with this modifier SHALL generate an error response.

7.7. Other Qualities

A Quality column defines other qualities not covered in other columns. Some qualities are limited to a specific set of elements. If an element does not have designated qualities, then it SHALL inherit qualities from the next highest element in the model hierarchy. For example: A data field in a struct attribute inherits its access qualities from attribute.